STK Asistanı — Assignment 2: Working Prototype and Model Development

AI for the SDGs · Capstone · Group 6: Ufuk Gündoğar, Tuna Bilgen · 30 October 2026

GitHub repository (code, data, README): https://github.com/tunabilgen/stk-asistani
Screen recording (2:38): https://drive.google.com/file/d/1lPLXOffwEZDrbJRBoWqMd4cP7gibOF5S/view?usp=sharing
Run it: follow the README section "Run it" (install, download data, train models, build, `python app.py`). A fresh clone already shows the map, campaign drafts and saved province profiles for the top-ranked countries.

1) What the prototype does, end to end

STK Asistanı helps NGOs decide **where** to run their next aid campaign, **what** it should contain and **with whom** to partner, with a focus on crises that mainstream media ignore. The working prototype runs the full chain on real, current data:

1. **Ingest** independent sources (UCDP conflict deaths, Global Hunger Index, IFRC appeals, Tech for Palestine, NASA EONET, USGS), 26 news feeds including 8 Turkish outlets, and 30 days of **GDELT** media data (274,838 articles, 7,317 outlets).
2. **Classify** every news item into 10 humanitarian sectors with our trained model; Turkish and Arabic text is routed to a multilingual model.
3. **Score** every country: need, trend, aid gap and the **need-relative silence index** (section 4).
4. **Show** the result on an interactive map with a country file (score breakdown, trend, sources, verified Red Crescent / Red Cross partner, province-level project ideas).
5. **Draft** a campaign with an LLM (DeepSeek V4.1 Flash) that may use only the verified facts; a validator checks every number and partner name, and a human approves the text.

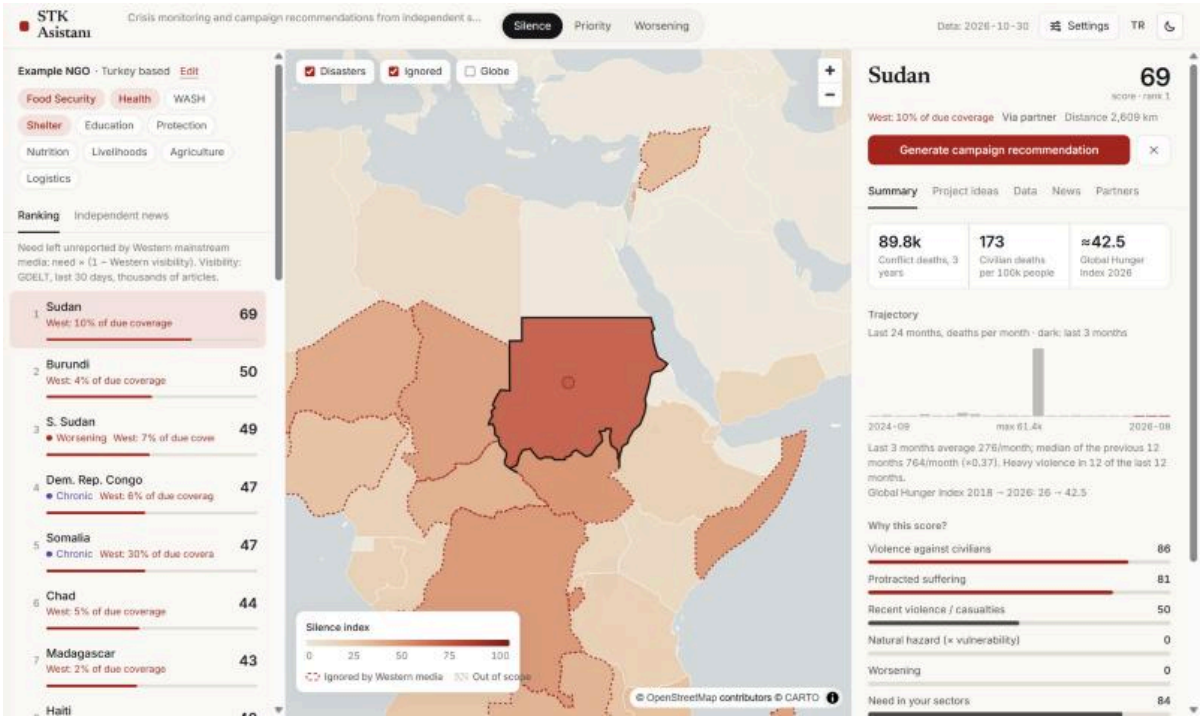


Fig. 1 — The prototype opens on the Silence view. Sudan is first: very high need, and Western media give it 10% of the coverage its need would justify.

2) Models tried for the sector classifier

Task and data. Multi-label classification of humanitarian text into 10 sectors (food, health, WASH, shelter, education, protection, nutrition, livelihoods, agriculture, logistics). Training data: HumSet (Fekih et al., 2022), expert-labelled excerpts in English, French and Spanish. All models are tested on the same official test split of **12,384 excerpts**, and every learned model gets per-class decision thresholds tuned on the validation split, so the comparison is fair.

Metrics. One text can have several sectors, so accuracy is not meaningful. We use **F1**, the balance of precision and recall. **Micro-F1** pools all decisions, so frequent sectors dominate it; **macro-F1** averages the 10 sectors equally and shows whether rare sectors (logistics, agriculture) also work. We also report F1 per language.

#	Approach	micro-F1	macro-F1	Note
1	Keyword rules	0.558	0.539	no learning, baseline
2	TF-IDF + Complement Naive Bayes	0.677	0.631	
3	TF-IDF + linear SVM	**0.781**	**0.745**	best, but no probabilities
4	TF-IDF + logistic regression	0.780	0.740	**used in the app** for EN/FR/ES
5	Multilingual sentence embeddings (frozen MiniLM) + LogReg	0.737	0.696	**used in the app** for TR/AR
6	Fine-tuned multilingual MiniLM transformer	0.413	0.422	laptop CPU: 12k excerpts, 2 epochs, 73 min
7	LLM zero-shot, DeepSeek V4.1 Flash	0.818	0.717	preliminary: 37 excerpts; model 4 on the same 37: 0.718 / 0.594

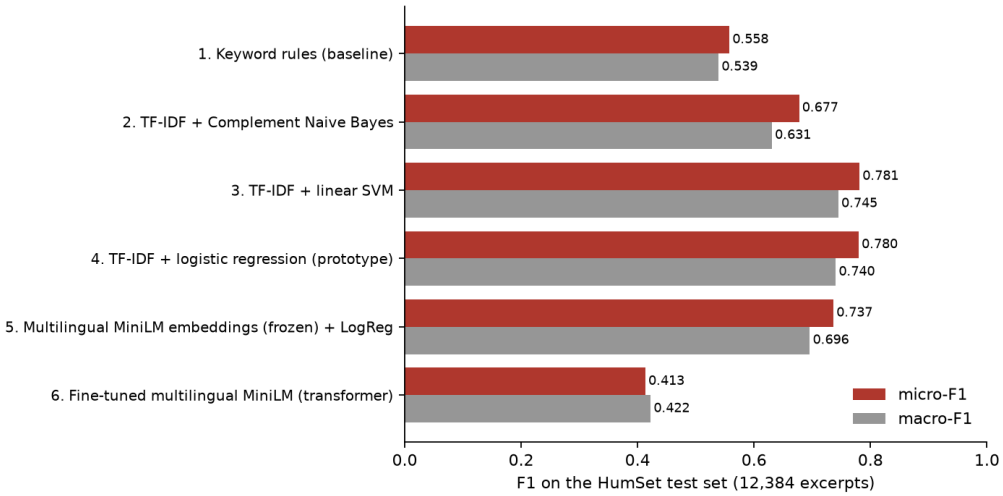


Fig. 2 — Models 1–6 on the full HumSet test set (12,384 excerpts).

Which performed better and why. The linear models (SVM 0.781, logistic regression 0.780) were best on the full test set. HumSet excerpts are long and full of sector words (*cholera, latrine, school*), so word n-grams carry most of the signal, and with 100k+ labelled examples a linear model learns them well and retrains in about a minute, which our self-training loop needs. We keep logistic regression instead of SVM because it gives probabilities, which the dashboard uses. The frozen sentence-embedding model is about 4 points lower because it compresses each text into one 384-number vector. The fine-tuned transformer did badly **because of our compute budget, not the method**: on a laptop CPU we could train on only 10% of the data for 2 epochs, and the loss was still falling; we report this negative result instead of hiding it. The LLM, with no training at all, beat our model by about 10 points on the same 37 excerpts, but the free model hit its daily usage limit after 37 calls, so this is preliminary; the run resumes from its cache and the full 200 will be reported in Assignment 3. Even if the LLM is more accurate, it is too slow and rate-limited to label thousands of news items a day, so the app uses it only for writing campaign drafts, where a human checks the output.

3) Turkish and Arabic

HumSet has no Turkish or Arabic text, but our users read Turkish media and many crises are reported in Arabic. We machine-translated 300 English test excerpts with open OPUS-MT models (labels carry over; the excerpts were never seen in training) and scored every model in all three languages.

micro-F1	English	Turkish	Arabic
Keyword rules	0.599	0.151	0.047
TF-IDF + LogReg (Assignment 1 model)	**0.792**	0.336	0.200
Multilingual embeddings + LogReg	0.732	**0.699**	**0.704**
App: language router	**0.792**	**0.695**	**0.704**

The Assignment 1 model **could not read Turkish or Arabic**. The app now detects the language of each text and sends EN/FR/ES to TF-IDF and Turkish/Arabic to the multilingual model, so it keeps the best score in every language. Limitation: machine translations are cleaner than real news; a small hand-labelled set of real Turkish and Arabic headlines is planned for Assignment 3.

4) The silence idea at the centre: need-relative silence

Assignment 1 used *need × (1 − attention)*, which treats ten articles about a catastrophe the same as ten articles about a small crisis. Now each crisis is compared with the coverage its need would justify:

```
coverage ratio = (country's share of Western foreign news) ÷ (country's share of total need)
silence = need × (1 − min(coverage ratio, 1))
```

A ratio of 1 means fair coverage; 0.1 means the crisis gets 10% of it. Home-country news (USA, UK) is excluded. The interface shows the ratio ("West: 10% of due coverage") instead of a yes/no "ignored" flag.

Top of the ranking: **Sudan** (10% of due coverage), Burundi, South Sudan, DR Congo, Somalia, Chad, Madagascar, Niger, Haiti, Timor-Leste. Palestine receives 59% of its due coverage, Ukraine 346%. Attention now comes from GDELT instead of ~90 RSS items. The two samples agree well (attention $\rho = 0.66$; silence ranking $\rho = 0.80$; 8 of the top 10 identical), so the Assignment 1 finding holds.

5) Media attention vs need vs funding (GDELT, Turkish media, OCHA FTS)

- **Attention follows need only weakly.** Across 35 high-need countries, Spearman $\rho = 0.25$ for Western media, 0.29 for Turkish media and 0.14 for all GDELT outlets (none significant at 5%).
- **Turkish vs Western media.** Turkish outlets give 33% of their foreign news to these crises, Western outlets 19% (Fig. 3).
- **UN funding as a benchmark only.** OCHA FTS data is never used in the ranking, so the ranking stays independent. Funding per person follows need ($\rho = 0.59$). Controlling for need, more Western attention goes with somewhat more funding (partial $\rho = 0.28$, $p = 0.10$). The quiet half of crises receives a median \$3.9 per person, the visible half \$51.6, but the visible half also has higher need, so this is a hint, not proof.
- **Where is Gaza?** Of stories whose web address names Gaza or Palestine, GDELT's geocoder placed 90% (Western), 96% (Turkish) and 90% (other outlets) **only in Israel**. Because the rate is similar for all groups, this is mainly a geocoder effect, not one group's bias, but any tool that trusts GDELT's locations will under-count Palestine. We re-attribute stories by the place named in the address, with the same rule for every country.

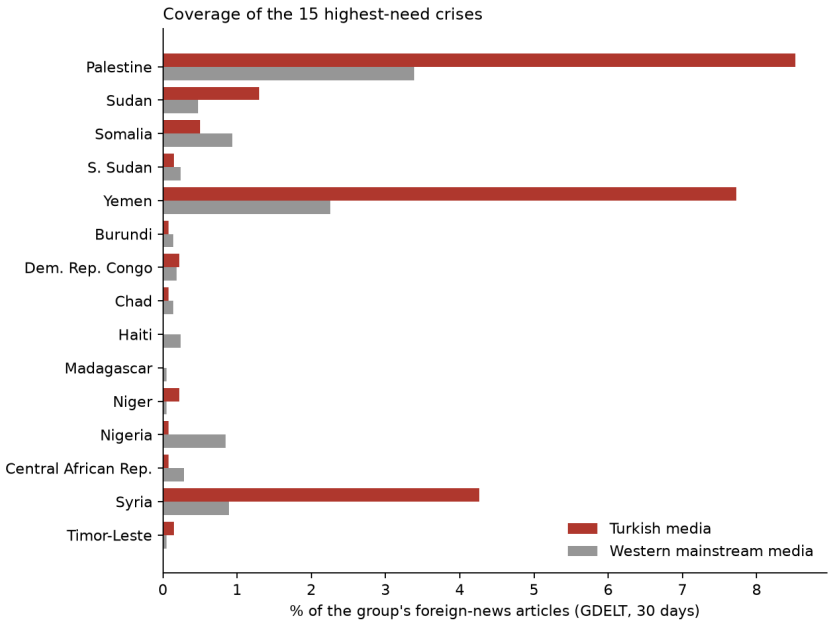


Fig. 3 — Share of each media group's foreign news given to the 15 highest-need crises (GDELT, 30 days).

6) LLM integration and safety

Campaign and project drafts are written by **DeepSeek V4.1 Flash**, called through the official OpenCode command-line tool, so the repository contains no API key. The prompt allows only the facts we pass (numbers with named sources, verified partner candidates). A validator then removes partner names that are not in our candidate list and flags any number that does not appear in the facts. A human must approve every text. The free model has a daily usage limit; when it is unreachable the app shows the last saved draft for that country, **labelled with the model that wrote it** and re-validated against today's data, or a template. During development the free quota ran out: the Sudan (Turkish) draft was written by DeepSeek, and five more drafts were written by Claude Opus 5.5 using the same prompt and facts. Each one is labelled as such in the app and in the README.

7) Changes made after the Assignment 1 feedback

Feedback	What we did
Put the silence idea at the centre; simplify the rest	The app opens on the Silence view; silence is now relative to need; the yes/no flag became a coverage ratio.
Use GDELT	Western attention now comes from GDELT 2.0 (274,838 articles, 7,317 outlets, 30 days).
Use ReliefWeb and OCHA FTS	OCHA FTS added as a benchmark only (“does UN money reach the crises the media ignore?”), never in the ranking. ReliefWeb is not used: it is also run by UN OCHA, and the ranking deliberately stays independent of UN data; it may be added as a second benchmark in Assignment 3.
Add Turkish media; compare with Western media	8 Turkish outlets across the political spectrum (TRT Haber, AA, Hürriyet, Yeni Şafak, Sabah, Cumhuriyet, BirGün, bianet) plus Turkish GDELT outlets; comparison in section 5.
Test the classifier on Turkish and Arabic	New EN/TR/AR test set; language router in the app (section 3).
Interview 2–3 NGOs	We could not arrange interviews in this period. In Assignment 3 we replace them with structured persona tests (small Turkish NGO, international NGO, journalist, donor) and state clearly that this is a weaker form of user validation.
Discuss ranking sensitivity	First step done (GDELT vs RSS check, section 4); a full sensitivity analysis of the weights is planned for the Assignment 3 report.

8) Known limitations

- The list of “Western mainstream” outlets (24 domains) is our choice; a different list could change individual ratios.
- Turkish and Arabic scores come from machine translations, which are cleaner than real news.
- GDELT's geocoder misplaces Palestine stories; our URL-based correction fixes only stories whose address names the place.
- The fine-tuned transformer was limited by CPU compute; the LLM evaluation covers 37 of 200 planned excerpts.
- No real NGO users have tested the tool yet.

9) Plan for Assignment 3

Work package	What we will do	Output
Framing analysis (the core research question)	Beyond *how much* media cover a crisis, measure *how*: are victims named or only counted? Active ("were killed by") or passive ("died") voice? Who is quoted: officials, aid agencies or affected people? An LLM labels headlines from Western, Turkish and other outlets; we check it against ~100 hand-labelled headlines before trusting it.	framing scores per outlet group and crisis; accuracy of the LLM labeller
Persona tests (instead of NGO interviews)	Run fixed scenarios as four users: a small Turkish NGO (health, low budget), an international NGO, a journalist looking for under-reported stories, a donor. Record whether the ranking, country file and campaign draft are useful, correct and understandable for each.	test protocol and results table
Error and bias audit	Test with varied inputs (languages, sectors, excluded countries, missing data) and document every wrong, inconsistent, biased or misleading output, its cause and the fix.	error log with before/after
Ranking sensitivity	Change each weight and the outlet list within reasonable ranges and measure how stable the top 10 is (rank correlation, overlap).	sensitivity figure in the report
Model evaluation	Finish the 200-excerpt LLM comparison; label ~100 real Turkish and Arabic headlines to check the language router on real text.	updated tables
Final deliverables	Polished repository and README, 3–5 page final report, English presentation and a ~5-minute video.	submission